Carol Hanna

London, United Kingdom

Computer Scientist

✉ carol.hanna@ed.ac.uk

CV - November 2026

Research Interests: Automated Software Repair, Hot Fixing in Production Systems, Genetic Improvement, AI-driven Software Engineering, Software Reliability.

EDUCATION

PhD - ***Computer Science*** **-** University College London, UK ***Jan. 2023 - Jun. 2026***

Pass without corrections

Doctoral Dissertation: Fixing Under Fire: Practices, Perspectives, and Automation for Hot Fixing Time-Critical Software Issues

MSc - ***Software Systems Engineering*** **-** University College London, UK ***Sep. 2021 - Sep. 2022***

Distinction

Master's Dissertation: Optimising mutation operator selection for search-based automated program repair with reinforcement learning.

BSc - ***Computer Science*** **-** Technion, Israel ***Oct. 2016 - Oct. 2020***

Undergraduate Dissertation: Binary2Name- Automatic detection of the functionality of a binary code snippet using a deep neural network. Achieved excellent results (above 60% F1 value).

EMPLOYMENT

Lecturer/Assistant Professor of Software Engineering, **University of Edinburgh**, UK ***Nov. 2026***

Research-track academic in the School of Informatics.

Principal Research Scientist, **Code SA,** UK ***Sep. 2025 - Feb. 2026***

I lead research on improving the automated code analysis and test generation services by identifying inefficiencies and reliability gaps and developing solutions that drive impact.

Teaching Assistant and Guest Lecturer, **University College London**, UK ***Sep. 2023 - Dec. 2025***

Guest lectures, leading tutorials and grading coursework. Modules include: Research Methods, Algorithms I, and Automated Software Engineering.

Venture Fellow, **Creator Fund**, United Kingdom ***Oct. 2024 - Oct. 2025***

Sourced and evaluated AI-focused deals for a pre-seed fund investing in university-founded startups; advised portfolio companies on product strategy and growth post-investment.

Research Assistant, **University College London**, United Kingdom ***Oct. 2022 - Jan. 2023***

RA at the Software Systems Engineering group at UCL, with research placement resulting in a journal publication.

Verification Engineer, **Intel Corporation**, Israel ***Aug. 2019 - Jul. 2021***

Created functional verification environments for several components of the Mobileye sixth-generation autonomous driving chip architecture, including the direct memory access unit, the scalar processor, and an accelerator on the chip. Promoted after 8 months at the company, while still a full-time student.

Student Mentor, **Technion**, Israel ***Sep. 2018 - Jul. 2019***

Adviser for incoming freshmen and students who are struggling academically.

Research Assistant, **Technion**, Israel ***Jun. 2018 - Oct. 2018***

Worked closely with the Dean of the Education, Science, and Technology faculty on research that focused on migrating the portfolios and work of students to the digital world.

PUBLICATIONS

Citations: 800< ; h-index: 8 (Google Scholar, July 2026)

Under review at top-tier venues:

1. Fast First, Flawless Later: Re-evaluating Automated Program Repair for Real-World Hot Fixing
2. Improving the Improvement: Surrogate-Guided Test Execution in Search-Based Software Engineering
3. ReGenSQL: A Hybrid Query Rewrite Framework with Genetic Improvement and Large Language Models
4. Reinforcement Learning for Context-Aware Mutation Selection in LLM-Based Genetic Improvement

Conferences:

5. Hot Fixing in the Wild - 2026 - EASE.
6. Behind the Hot Fix: Demystifying Hot Fixing Industrial Practices at Zühlke and Beyond - 2025 - FSE.
7. LLM-Guided Genetic Improvement: Envisioning Semantic Aware Automated Software Evolution - 2025 - ASE
8. Optimised fitness functions for automated improvement of software's runtime - 2025 - SSBSE.
9. Toward Systematizing Hot Fixing for Production Software - 2024 - FSE.
10. Exploring LLM-driven explanations for quantum algorithms - 2024 - ESEM.
11. Hot Patching Hot Fixes: Reflection and Perspectives - 2023 - ASE.
12. Evaluating Explanations for Software Patches Generated by Large Language Models - 2023 - SSBSE.
13. Enhancing Genetic Improvement Mutations Using Large Language Models - 2023 - SSBSE.

Journals:

14. A Mixed-Method Study of Hot Fixing in Industry: Practices, Bottlenecks, and Opportunities - 2026 - TOSEM.
15. Hot Fixing Software: A Comprehensive Review of Terminology, Techniques, and Applications - 2025 - TOSEM.
16. Reinforcement Learning for Mutation Operator Selection in Automated Program Repair - 2025 - ASE Journal.
17. Large Language Model-Based Mutations in Genetic Improvement - 2024 - ASE Journal.

Workshops:

18. Improving a Parallel C++ Intel SSE SIMD Linear Genetic Programming Interpreter - 2026 - GI Workshop @ ICSE.
19. Applying Genetic Improvement Techniques for Automated Program Repair of Transpiled Code - 2026 - GI Workshop @ ICSE.
20. From Kotlin to Swift and Back: Toward Fully Automated Cross-Language Code Transpilation - 2025 - A-Mobile.
21. Large Language Model-based Code Completion is an Effective Genetic Improvement Mutation - 2025 - GI Workshop @ ICSE.
22. Enhancing Software Runtime with Reinforcement Learning-Driven Mutation Operator Selection in Genetic Improvement - 2025 - GI Workshop @ ICSE.
23. An Analysis of the Automatic Bug Fixing Performance of ChatGPT - 2023 - APR Workshop.

AWARDS

- UCL computer science departmental “Best Thesis” award (runner up) 2026.
- Genetic Improvement Workshop “Best Paper” award for “Large Language Model based Code Completion is an Effective Genetic Improvement Mutation” 2025.
- UCL departmental PhD scholarship: competitive scholarship supporting doctoral research 2023-2027.

PROFESSIONAL SERVICE

Leadership Roles:

- Software Testing, Verification and Reliability journal (STVR) associate editor.
- SSBSE conference steering committee member.
- Track co-chair of SSBSE Challenge Track 2025 and 2026.

- Organizer of the bi-weekly Software Systems Engineering Group Invited Speakers Seminar Series.
- Co-organizer of the 67th, 68th, and 70th CREST open workshops.

Supervisory Roles:

- Co-supervisor for Master's student dissertations:
    - Hassan Kandil 2025.
    - Devu Raju Kurup 2025.
    - Jingyuan Wang 2024.
    - Dimitris Bouras 2024.
    - Hancheng Zuo 2024.
- Co-supervisor for Undergraduate student dissertations:
    - Levon Davtyan 2026.
    - Prasham Jadhwani 2024.
    - Sachi Lad 2024.
    - Damien Bose 2024.

Programme Committees & Reviewing:

- Peer reviewer for ACM Transactions on Software Engineering and Methodology (TOSEM) journal.
- Peer reviewer for IEEE Transactions on Software Engineering (TSE) journal.
- Peer reviewer for Empirical Software Engineering (EMSE) journal.
- Peer reviewer for Automated Software Engineering (ASE) journal.
- Peer reviewer for IEEE Transactions on Reliability journal.
- Peer reviewer for Artificial Intelligence Review journal.
- Peer reviewer for Software and Systems Modeling journal.
- Peer reviewer for Software: Evolution and Process journal.
- ICST research track 2027 program committee member.
- ICPC early research achievements track 2027 program committee member.
- ASE research track 2026 program committee member.
- FSE artifacts track 2026 program committee member.
- ASE tool demonstration 2025 and 2026 program committee member.
- SSBSE main Track 2025 and 2026 program committee member.
- Evaluation of Qualitative Aspects of Intelligent Software Assistants (EQUISA) 2026 program committee member.
- MSR main Track 2025 program committee member.
- Interware tool demonstration 2025 program committee member.
- LLM4Sec 2025 program committee member.
- Genetic Improvement workshop 2024 and 2025 program committee member.
- ICTSS challenge 2024 program committee member.
- SSBSE RENE/NIER 2024 program committee member.

Volunteering & Outreach:

- Student volunteer at ICSE 2025; FSE 2025; ASE 2025.
- Session chair: ASE 2023; ASE 2025; COW 65 and 67; bi-weekly SSE seminar.
- Photography and social media for the 66th UCL CREST Open Workshop.
- Participation in ASE 2023, FSE 2024, ICSE 2025, FSE 2025, ASE 2025, EASE 2026.

- Volunteered to develop the official Android application for the Society for the Prevention of Cruelty to Animals in my local community. Published in the Google Play Store.

TALKS, LECTURES, AND EVENTS

- Invited to Dagstuhl Seminar 26372 “Resilient AI-Driven Software Systems”.
- Invited talk at the 71st CREST Open Workshop “Agentic Testing and Verification”.
- EASE 2026 paper presentation.
- Invited talk at King’s College London SSY Group Seminar.
- ASE 2025 paper presentations.
- SSBSE 2025 paper presentation.
- A-mobile 2025 paper presentation.
- Invited talk at the 67th CREST Open Workshop “AI-powered advances in genetic improvement”.
- FSE 2025 paper presentation.
- GI workshop@ICSE 2025 paper presentation.
- Guest Lecture in ASE module at UCL, Autumn 2024 and Autumn 2025.
- FSE 2024 paper presentation.
- ASE 2023 paper presentation.
- UCL SSE seminar 2022 on applying reinforcement learning for automated program repair.

PROGRAMMING: C, C++, Java, Python, Verilog | **LANGUAGES**: English, Arabic, Hebrew (all fluent)